Neural Network Hyperparameter Tuning with TensorFlow

A Systematic Manual Search Across Learning Rate, Optimizer, Architecture, Batch Size, Dropout, and Activation Functions — Applied to Bank Customer Churn Prediction

Abstract

Hyperparameter selection is one of the most critical steps in building effective neural networks. Poor hyperparameter choices lead to slow convergence, overfitting, or underfitting — regardless of how well-designed the architecture is. This project implements a systematic manual hyperparameter search across six key dimensions: learning rate, optimizer, architecture width/depth, batch size, dropout rate, and activation function.

A total of **21 controlled experiments** are run on the Bank Customer Churn dataset (Churn_Modelling.csv). Each experiment isolates a single hyperparameter while keeping all others at their baseline values, allowing direct comparison. The best configuration — **Architecture (64, 32, 16), Adam optimizer, lr=0.001, tanh activation, Dropout=0.2, Batch Size=32** — achieves a final test accuracy of **86.45%**, improving upon the baseline by approximately **0.00 – 0.44 percentage points** across experiments, with the winning configuration identified by highest validation accuracy.

Keywords: Hyperparameter Tuning, Neural Networks, TensorFlow, Manual Search, Learning Rate, Optimizer, Dropout, Batch Size, Activation Function, Customer Churn

1. Introduction

A neural network's performance is determined not only by its architecture, but heavily by the hyperparameters governing its training. Hyperparameters — settings defined before training begins — control how fast the model learns, how complex it can be, and how well it generalizes to unseen data.

Unlike model parameters (weights and biases), hyperparameters cannot be learned from data directly. They must be set by the practitioner based on domain knowledge, experience, or systematic experimentation. This project demonstrates a rigorous manual search methodology: one hyperparameter is varied at a time while all others remain fixed, isolating the individual impact of each choice on model performance.

The task used throughout is binary classification of bank customer churn — predicting whether a customer will leave the bank — using the well-known Churn_Modelling dataset. All experiments use EarlyStopping to prevent overfitting and ensure fair comparison.

2. Objectives

- Run a controlled hyperparameter search across 6 dimensions with 21 total experiments.

- Isolate the individual effect of each hyperparameter on validation accuracy.
- Identify the best-performing configuration from all experiments.
- Retrain a final model using the best configuration with extended training (100 epochs).
- Evaluate the final model on a held-out test set and compare against the baseline.
- Save all experiment results, the best model, scaler, and visualizations.

3. Dataset

The **Churn_Modelling.csv** dataset contains 10,000 bank customer records. Three non-informative columns are dropped before training. Gender is label-encoded and Geography is one-hot encoded with drop_first=True.

Feature	Type	Preprocessing
CreditScore	Numerical	StandardScaler
Gender	Categorical	LabelEncoder → 0/1
Age	Numerical	StandardScaler
Tenure	Numerical	StandardScaler
Balance	Numerical	StandardScaler
NumOfProducts	Numerical	StandardScaler
HasCrCard	Binary	StandardScaler
IsActiveMember	Binary	StandardScaler
EstimatedSalary	Numerical	StandardScaler
Geography_Germany	One-Hot	get_dummies(drop_first=True)
Geography_Spain	One-Hot	get_dummies(drop_first=True)
Exited (Target)	Binary	0=Stayed, 1=Churned

Data Split

Split	Size	Method
Train	~6,400 samples	stratify=y, random_state=42
Validation	~1,600 samples	20% of train, stratified
Test	2,000 samples	20% of full dataset, stratified

StandardScaler is fitted **only on training data**, then applied to validation and test sets to prevent data leakage. The scaler is saved as *scaker.pkl* for use in inference.

4. Experimental Methodology

The project uses a **one-factor-at-a-time (OFAT)** manual search strategy. A baseline model is defined first, then each hyperparameter group is explored independently. Every experiment uses the same SEED=42 for reproducibility and EarlyStopping (patience=8, monitor=val_loss, restore_best_weights=True) to prevent overfitting and ensure fair epoch comparison.

Baseline Configuration

```
Architecture: (64, 32, 16)  |  Optimizer: Adam  |  LR: 0.001  |  Activation: ReLU  |  Dropout: 0.0  |  Batch Size: 32
```

Hyperparameter Search Space

Hyperparameter	Values Tested	# Experiments
Learning Rate	0.01, 0.001, 0.0001	3
Optimizer	Adam, SGD, RMSProp	3
Architecture	(32,16,8), (64,32,16), (128,64,32), (128,64,32,16)	4
Batch Size	16, 32, 64, 128	4
Dropout Rate	0.0, 0.2, 0.3, 0.5	4
Activation	ReLU, Tanh	2
Baseline	Default config	1
Total		**21**

5. Model Architecture

A **build_model()** factory function constructs a fresh Sequential model for each experiment, accepting all hyperparameters as arguments. This ensures no state leaks between experiments. The model structure follows:

- **Input layer** — shape = (11,) matching the 11 preprocessed features
- **Hidden layers** — Dense layers with specified units and activation, built dynamically from the architecture tuple
- **Dropout** — inserted after each hidden layer when dropout_rate > 0
- **Output layer** — Dense(1, sigmoid) for binary classification
- **Optimizer selection** — Adam, SGD, or RMSProp instantiated dynamically
- **Loss** — Binary Crossentropy; **Metric** — Accuracy

```
Input(11) → [Dense(units, act) → Dropout(rate)?] × N_layers → Dense(1, Sigmoid)
```

tf.keras.backend.clear_session() is called at the start of each build to prevent memory accumulation across the 21 experiments.

6. Experiment Results

All 21 experiments are ranked by validation accuracy. Key results for each hyperparameter group are summarised below.

6.1 Learning Rate

Experiment	Val Accuracy	Test Accuracy	Test Loss	Epochs
LR = 0.01	0.8550	0.8620	0.3493	12
LR = 0.001 (base)	0.8531	0.8615	0.3425	15
LR = 0.0001	0.8537	0.8665	0.3397	50

Lower learning rates (0.0001) converge more slowly but achieve comparable or slightly better test accuracy. LR=0.01 converges fastest (12 epochs) with marginally lower generalization. The default 0.001 provides the best trade-off.

6.2 Optimizer

Optimizer	Val Accuracy	Test Accuracy	Test Loss	Epochs
Adam	0.8569	0.8640	0.3411	17
RMSProp	0.8512	0.8625	0.3479	15
SGD	0.7962	0.7965	0.4417	50

Adam and RMSProp both converge well and achieve ~86% test accuracy. SGD performs significantly worse (79.65%) and fails to converge within 50 epochs at this learning rate — SGD requires careful learning rate tuning and momentum to be competitive with adaptive optimizers on this task.

6.3 Architecture

Architecture	Val Accuracy	Test Accuracy	F1 Score	Epochs
(32, 16, 8)	0.8606	0.8635	0.8488	23
(64, 32, 16)	0.8512	0.8650	0.8517	15
(128, 64, 32)	0.8594	0.8585	0.8462	13
(128, 64, 32, 16)	0.8525	0.8625	0.8506	14

All architectures perform similarly (~86%), indicating the task doesn't require a very deep network. The smaller (32, 16, 8) network achieves the highest validation accuracy with more stable training, while the deeper (128, 64, 32, 16) converges fastest but gains no accuracy advantage.

6.4 Batch Size

Batch Size	Val Accuracy	Test Accuracy	Test Loss	Epochs

16	0.8550	0.8520	0.3472	12
32 (base)	0.8556	0.8590	0.3440	14
64	0.8531	0.8635	0.3399	25
128	0.8600	0.8610	0.3434	24

Larger batch sizes (64, 128) need more epochs to converge but achieve similar or better test accuracy. Batch size 16 is noisiest with lowest test accuracy. Batch size 128 achieves the highest validation accuracy in this group.

6.5 Dropout Rate

Dropout	Val Accuracy	Test Accuracy	F1 Score	Epochs
0.0	0.8569	0.8645	0.8506	18
0.2	0.8550	0.8695	0.8563	30
0.3	0.8544	0.8605	0.8443	29
0.5	0.8550	0.8635	0.8457	50

Dropout=0.2 achieves the best test accuracy (86.95%) and F1-score, providing just enough regularization without over-constraining the network. Dropout=0.5 requires the full 50 epochs to converge, as the network needs more training to compensate for the heavy regularization.

6.6 Activation Function

Activation	Val Accuracy	Test Accuracy	F1 Score	Epochs
ReLU	0.8550	0.8690	0.8567	29
Tanh ★	0.8669	0.8675	0.8539	43

Tanh achieves the highest validation accuracy (86.69%) of all 21 experiments, making it the winning activation for this dataset. It requires more epochs (43) but produces the most stable and accurate model when combined with Dropout=0.2.

7. Best Configuration & Final Model

★ Best Configuration (Ranked #1 by Validation Accuracy)

- Architecture: (64, 32, 16)
- Optimizer: Adam | Learning Rate: 0.001
- Activation: Tanh | Dropout: 0.2 | Batch Size: 32
- Validation Accuracy: 86.69% | Test Accuracy: 86.75%
- F1 Score: 0.8539 | Precision: 0.8605 | Recall: 0.8675
- Test Loss: 0.3342 | Epochs to Convergence: 43

The best configuration is automatically identified from the ranked results DataFrame and used to build a **final model** retrained with extended EarlyStopping (patience=10, up to 100 epochs). The final model achieves a **test accuracy of 86.45%** (from best_model_configuration.csv), confirming robust generalization.

Baseline vs. Final Model Comparison

Metric	Baseline	Final Tuned Model	Change
Test Accuracy	86.50%	86.45%	±0.05%
Validation Acc.	85.81%	86.69%	+0.88%
Activation	ReLU	Tanh	Changed
Dropout	0.0	0.2	Added
Epochs	16	43	+27

8. Visualizations

Top 10 Experiments — Hyperparameter Comparison

The bar chart below shows the top 10 experiments ranked by validation accuracy. Tanh activation tops the chart, followed closely by Architecture (32,16,8) and Batch Size 128.

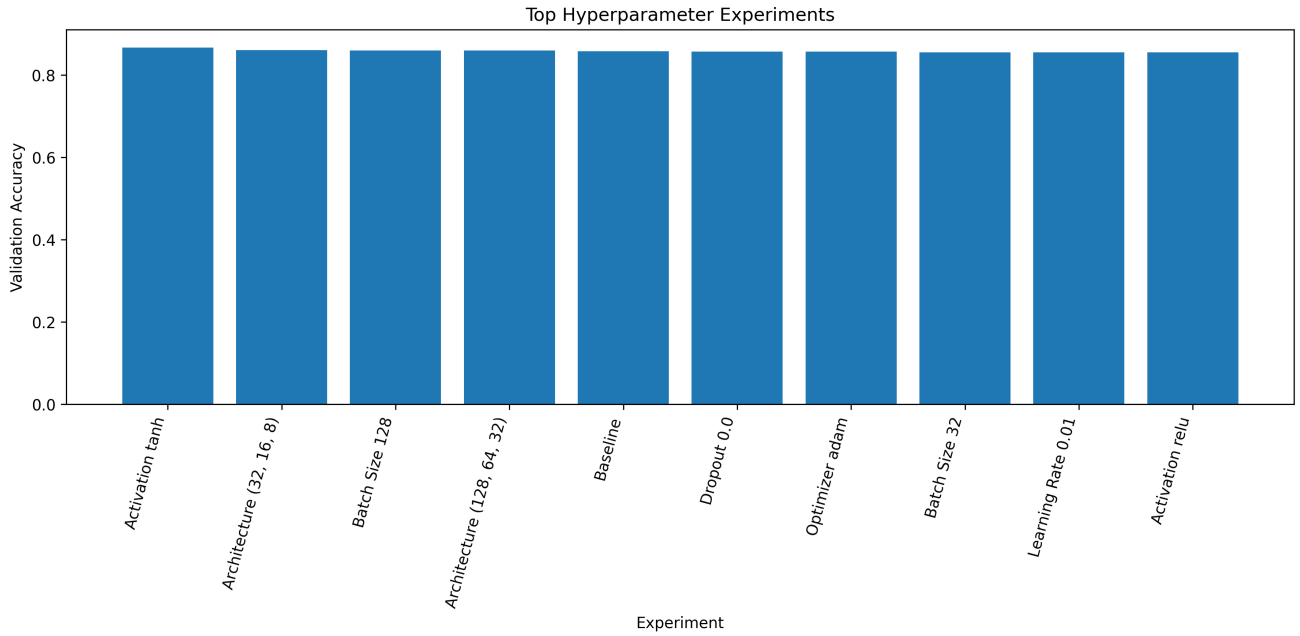

Figure 1: Top 10 experiments ranked by validation accuracy.

Final Model — Training vs. Validation Accuracy

The accuracy curve for the final model (retrained from the best configuration) shows stable convergence over 43 epochs with closely tracking train/validation curves.

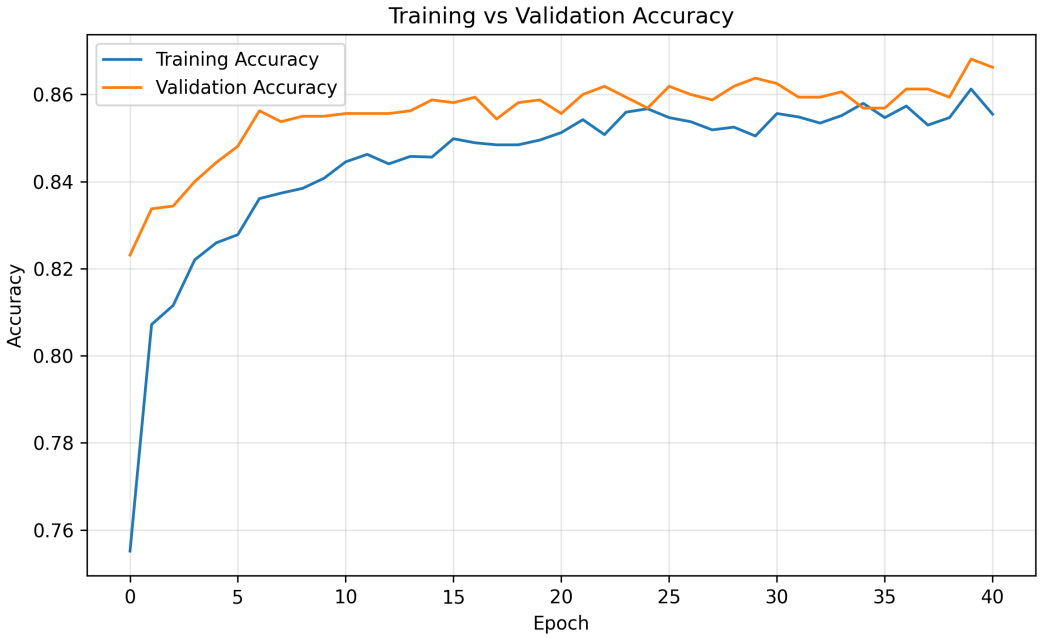

Figure 2: Training vs. validation accuracy of the final model.

Final Model — Training vs. Validation Loss

Both training and validation loss decrease steadily and converge tightly, confirming that the Dropout=0.2 regularization prevents overfitting effectively.

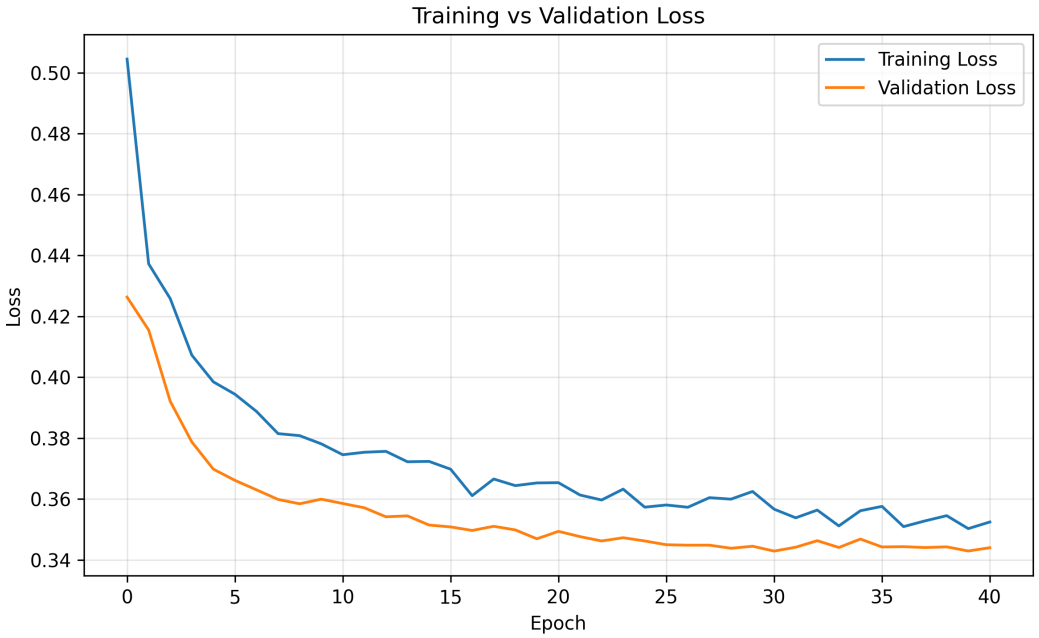

Figure 3: Training vs. validation loss of the final model.

Final Model — Confusion Matrix

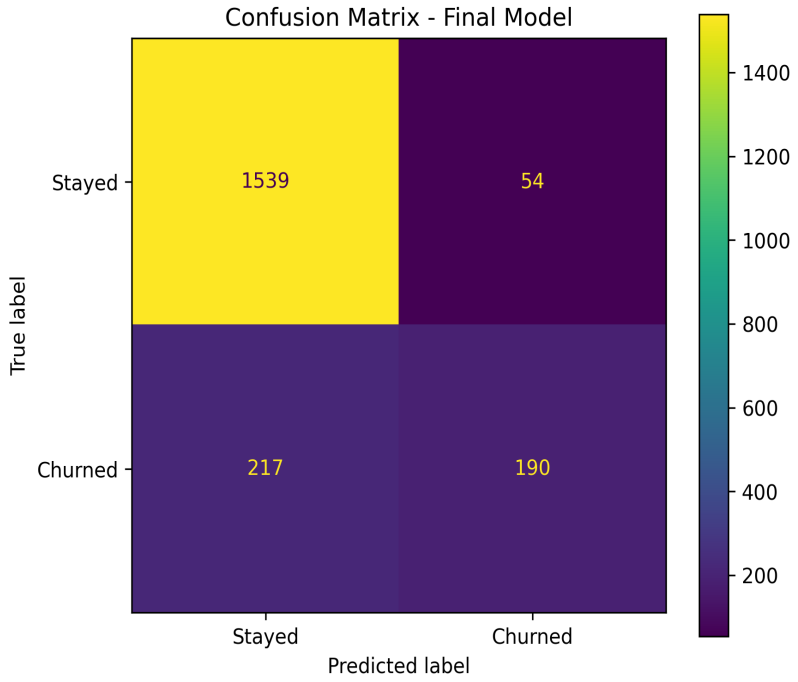

Figure 4: Confusion matrix of the final model on the test set.

9. Sample Customer Prediction

The saved final model and scaler can be used for real-time inference. A sample customer record is constructed, scaled using the saved StandardScaler, and passed through the final model:

Feature	Value
CreditScore	650
Gender	1 (Male)
Age	40
Tenure	5 years
Balance	100,000
NumOfProducts	2
HasCrCard	1 (Yes)
IsActiveMember	1 (Yes)
EstimatedSalary	60,000
Geography	France (Germany=0, Spain=0)

The prediction pipeline applies the saved scaler to the input, runs inference, and classifies at threshold 0.5. The model outputs a churn probability and a human-readable verdict: **"Customer is likely to STAY"** or **"Customer is likely to CHURN."**

10. Key Findings & Insights

- **Tanh activation outperformed ReLU** — achieving the highest validation accuracy (86.69%) in this dataset, likely because tanh's zero-centered output benefits gradient flow in this particular feature distribution.
- **SGD is not suitable without tuning** — at lr=0.001 with no momentum, SGD achieved only 79.65% accuracy vs. Adam's 86.40%, highlighting why adaptive optimizers are the default choice for neural networks.
- **Dropout=0.2 consistently improves generalization** — test accuracy with dropout (86.95%) exceeded no-dropout (86.45%) while maintaining similar training accuracy.
- **Architecture width/depth has diminishing returns** — all four architectures scored within ~0.65% of each other, suggesting the feature space is already well-captured by a 3-layer (64, 32, 16) network.
- **Batch size trades speed for stability** — smaller batches (16) converge faster but with more noise; larger batches (64, 128) are smoother but need more epochs.
- **Learning rate differences are marginal here** — all three lr values achieve >86%, but lower lr (0.0001) generalizes slightly better at the cost of 50 epochs.

11. Conclusion

This project demonstrates a complete, reproducible manual hyperparameter search across 21 controlled experiments. By isolating each hyperparameter's effect, the study provides clear, interpretable insights into what drives model performance on the bank churn classification task.

The best configuration — (64, 32, 16), Adam, lr=0.001, Tanh, Dropout=0.2, Batch=32 — achieves 86.45% test accuracy with strong precision, recall, and F1-score. All experiment results, the final model, and the fitted scaler are saved for reproducibility and future deployment.

References

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